Seat No : ________ No. of Printed Pages : 03

[12/A-6]

SARDAR PATEL UNIVERSITY, V.V.NAGAR

EXTERNAL EXAMINATION-2019

BCA Semester-I

Course Code: US01FBCA02

Course Title: Mathematics-I

Date: 10/02/2020 Monday

Time: 10.00 Am to 12-00 pm. Marks: 70

Q.1 MULTIPLE CHOICE QUESTIONS **[10]**

(1) Let A = { 1 , 2 , 3 , 4 } and B = { 2, 4 , 6 , 8 } then A U B = ……………………

(a) A U B = { 1 , 2 , 3 , 4 , 6 ,8 } (b) A U B = { 2 ,4 ,6 , 8 }
(c) A U B = { 2 ,4 , 6} (d) A U B = { 2 , 3 , 4 , 5 , 6}

(2) If U ={ 1 ,2 , 3 ,……………….., 9 } and A = { 1 , 2 , 3 , 4 }
Then A^c = ………………………

(a) { 1 , 2 , 3 , 4 } (b) { 5 , 6 , 7 , 8 , 9 } (c) { 3 , 4 , 5 , 6} (d) { 1 , 3 , 5 , 9 }

(3) A Π A =…………………

(a) Φ (b) A (c) U (d) None

(4) Let A ={ 0 , 1 } .Is a A closed under multiplication ?.............

(a) Yes (b) No (c) both (d) none

(5) Let D = { 2 , 4 , 6 ,……. } = { n : n is even }
Is D closed under : multiplication? addition? D is closed under………

(a) Multiplication (b) Addition (c) Both (d) None

(6) $(a^{-1})^{-1}$ = ……………

(a) E (b) a (c) none (d) a and b

(7) If u = (1,2) and v = (3,2) then u + v = ________

(a) (4,3) (b) (4,4) (c) (3,4) (d) (1,4) (e) none

(8) if u = (a, b) and v = (c, d) then u • v = ________

(a) (ab + cd) (b) (ac+bd) (c) ac +bd (d) ad + bc (e) none

(9) The arithmetic mean for simple frequency distribution is given by ________

(a) $\frac{\sum Xi}{n}$ (b) $\frac{\sum fixi}{n}$ (c) $\frac{\sum fixi}{N}$ (d) none

(10)__________ is the value dividing the data into two equal parts.

(a) mode (b) median (c) mean (d) none

Q.2 SHORT QUESTIONS (ANY TEN) **[20]**

(1) X= { red , blue }, Y={ blue, green , orange }, Z={ red , blue , white }
U={ red , yellow , blue , green , orange , purple , black , white }
Find (1) x^c ∩ (y\z) (2) $(x \cup y)^c$

(2) Determine the power set of A={ a , b , c , d }

(3) Define the terms with examples
(1) Complement of a set (2) Factorial function

(4) Define * on Q by a * b = a + b – ab. Is it associative? Justify Your answer

(5) Define Duality . Find duality of $(X \cup Y)^c = X^c \cup Y^c$

(6) Let F = { 2, 4 , 8 ,……. } = { x : x = 2^n , n ∈ N }.
Is F closed under : (a) multiplication ? (b) addition ?

(P.T.O)

(7) Consider u = (1,-2,3,4), v = (-2,4,5,-3) w= (5,-5,-3,4) and z = (2,7,4,-2).
Find the followings:
(i) u + v (ii) – 3u (iii) –w (iv) 3u – 4w

(8) Find x and y :
(i) if (x,7) = (4, x + y) (ii) if (20,y) = x (4,3)

(9) If u = (1,2,3,-4) , v = (5,-6,7,8) , w= (1,-2,5,3)
then find : (i) (u + v) • w (ii) u • v + v • w

(10) The intelligence quotients (IQ's) of 10 boys is given below:
70, 120, 110, 101, 88, 83, 95, 98, 107, 100 Find the Mean IQ

(11) Define Median with example

(12) Define Mode with example

Q.3 (A) By using Mathematical induction Method **[05]**
Prove that $1 + 3 + 5 + \dots\dots\dots + (2n - 1) = n^2$

(B) Find a formula for the inverse of f(x) = $\underline{(2x - 3)}$/5x - 8 **[05]**

OR

Q.3 (A) If $f(x) = x^2 - 3x + 2$ then find $f(x^2)$, f(2x-3), f(x + 2) **[05]**

(B) Find the power set of A={ { a , b } , { c } , { d , e , f } } **[05]**

Q.4 (A) Consider the set of rationals defined by a * b = a + b – ab **[05]**
Find (7 *1/2) Is (Q , +) semigroup? Is it commutative?

(B) Define f : R → R by f(x) =3x + 2 then find f^{-1} if exists **[05]**

OR

Q.4 (A) Let Q be the set of all rationals and * defined on Q by a * b = a + b – ab **[05]**
(1) Is(Q , *) a semi group?
(2) Find the identify element for *
(3) Find 3* 4 ,2 * (-5)

(B) Define : Group, subgroup , Group homomorphism and identity element of a group and prove it is unique **[05]**

Q.5 **[05]**

(A) Find the inverse of $A=\begin{pmatrix} 1 & 0 & 2 \\ 2 & -1 & 3 \\ 4 & 1 & 8 \end{pmatrix}$

(B) Find AB – BA where $A = \begin{pmatrix} 4 & 6 \\ -7 & 3 \end{pmatrix}$ and $B=\begin{pmatrix} 5 & -4 \\ -6 & 8 \end{pmatrix}$ **[05]**

OR

Q.5 (A) Solve by determinants : 2x - 3y = 7 3x + 5y = 1 **[05]**

(B) Find x and B if $B = \begin{pmatrix} 4 & x+2 \\ 2x-3 & x+1 \end{pmatrix}$ is symmetric **[05]**

Q.6 The following table gives the weights of 31 persons in a sample enquiry. Calculate mean, Harmonic mean, Geometric mean, Median and Mode. **[10]**

Weight	No. of Persons	Weight	No. of Persons
130	3	148	5
135	4	149	2
140	6	150	1
145	6	157	1
146	3		

OR

Q.6 Calculate the mean, median, mode, harmonic mean and geometric mean for the following data. **[10]**

xi	fi
10	14
30	23
50	27
70	21
90	15

—x—